

Suhani Shukla

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Education

University of California — Los Angeles (expected June 2027)

B.S. Computer Science, Minor in Entrepreneurship | GPA: 3.7/4.0

Relevant Coursework: Data Structures & Algorithms, Computer Organization, Operating Systems, Networking, Machine Learning, Deep Learning/Neural Networks, Natural Language Processing, Cybersecurity, Entrepreneurship & Venture Initiation

Honors and Awards: National Merit Scholar (2023), UCLA QWER Hacks 3rd Place (Traditional Hack) 2024, AIME Qualifier (2021)

Experience

Machine Learning Software Engineer Intern (Applied AI/ML), Microsoft AI

June 2026 - Present

- Designing and implementing end-to-end ML experimentation pipelines for teacher data generation and curation, running controlled ablations across teacher models, prompting strategies, and data recipes to optimize supervised fine-tuning (SFT) performance of Small Language Models (SLMs) powering Bing Generative Search.
- Training and evaluating trigger model classifiers to dynamically predict search fanout decisions, experimenting with query-text and post-retrieval search metadata signals to optimize routing between fanout and no-fanout queries.
- Improving reinforcement learning performance of student SLMs through reward-based skill optimization and iterative skill updates, improving policy learning and production model quality. Benchmarking and implementing SLM serving optimizations, including KV caching and quantization, to reduce inference latency and serving cost.
- Building automated evaluation pipelines to measure SLM quality across search tasks, analyzing model behavior and failure modes across training checkpoints to identify regressions and guide improvements to data and training recipes.

Software Development Engineer Intern, Amazon Web Services

June - Sept. 2025

- Automated end-to-end CI/CD workflows for AWS Redshift Console deployments using Live Pipeline Templates (Ruby) and AWS CDK (TypeScript), migrating legacy infrastructure to reproducible Infrastructure as Code (IaC).
- Onboarded services onto Apollo containers and Application Revisions, enabling deterministic builds and standardized releases across environments. Designed a zero-touch rollback system with real-time canary analysis and composite alarms, minimizing blast radius and ensuring safe global deployments across 20+ AWS regions.
- Reduced pipeline freshness from 100+ to <28 days and increased deployment frequency 4x through staged bake times, automated approvals, and promotion blockers. Integrated automated testing, performance monitoring, and deployment safety checks into release pipelines, improving reliability for thousands of enterprise customers globally.

Software Diagnostics Intern, Nokia

June - Aug. 2024

- Developed C++ systems-level diagnostic software for ASIC components, improving reliability of hardware workflows.
 - Implemented bidirectional gRPC streaming for inter-process communication in C++, enabling real-time system-wide logging and diagnostics across all *Diags* test suites. Ensured robustness under prolonged ASIC validation runs.
 - Designed and implemented custom POSIX signal handler to safely manage interrupts (Ctrl-C) for long-running diagnostics. Integrated with Python front-end for improved usability for test engineers and safer orchestration.
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Projects

HR AI Assistant (Production RAG Chatbot)

- Built and deployed a production internal HR chatbot for AlphaCore Wealth Advisory powered by Amazon Bedrock.
- Implemented backend services with FastAPI, FAISS, and embeddings-based retrieval, deployed serverless infra on AWS Lambda and S3. Built front-end interface, owning full-stack integration between UI and backend APIs.

EMG-to-QWERTY Keystroke Prediction (Deep Learning Project) 

- Evaluated CNN, LSTM/GRU, Transformer, and CNN-Transformer hybrid models for EMG-based keystroke prediction.
- Applied data augmentation techniques (e.g. Gaussian noise injection, random masking, dropout) and hyperparameter tuning; benchmarked models using Character Error Rate. Analyzed convergence behavior and overfitting trade-offs.

Easy EBT: Grocery Store EBT Item Finder 

- Ideated and created fully functional website to find and purchase EBT-eligible items at grocery stores using Python web scraping and recipe recommendations, with React JS and Django. Third place at UCLA QWER Hacks.
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Skills

Technical

Languages: C++, C, Python, TypeScript, JavaScript, Bash

AI/ML: PyTorch, HuggingFace, LangChain, LLMs, Small Language Models (SLMs), Reinforcement Learning, Supervised Fine-Tuning (SFT), RAG, Embeddings, CNNs, Model Evaluation, Amazon Bedrock

Web & Full-Stack: React, HTML/CSS, FastAPI, Django, MongoDB, Tailwind CSS/Material UI, Docker

Systems & Networking: Linux, Systems Programming, gRPCs, CI/CD, Infrastructure as Code (IaC)

Other

Quick learner, Excellent Communication, Fluency in Hindi and French